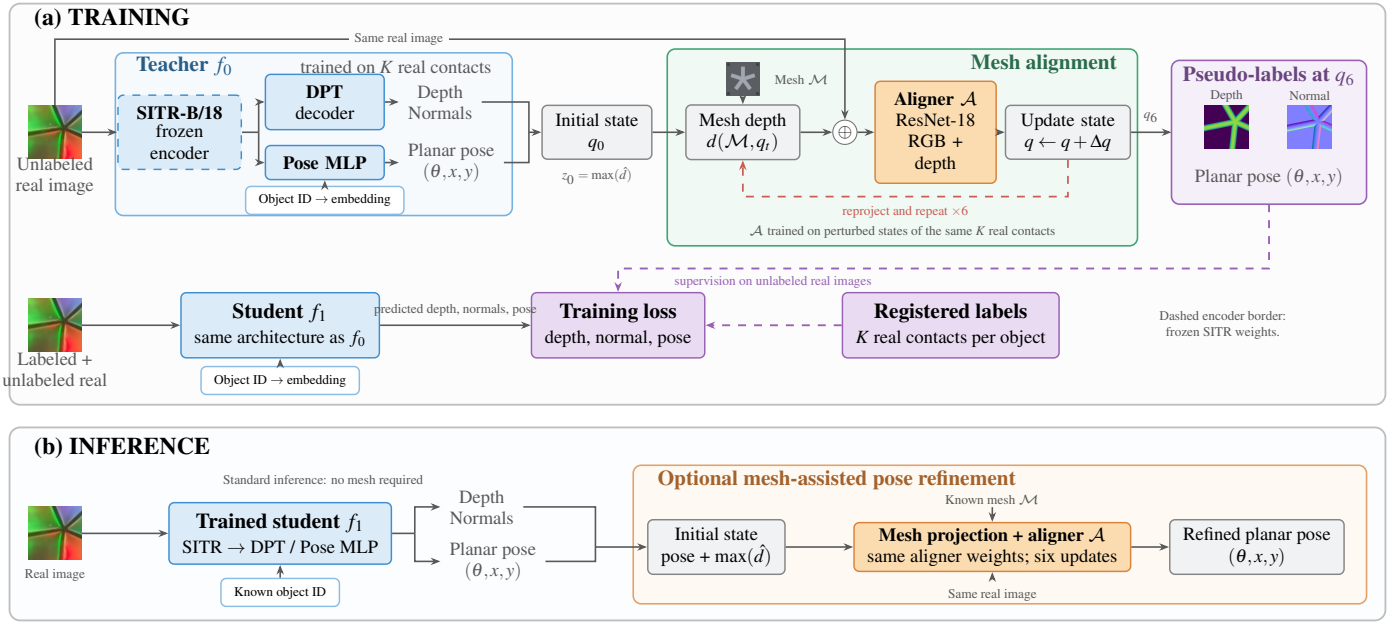

(a) TRAINING
Same real image
Teacher f_0
trained on K real contacts
SITR-B/18
frozen
encoder
DPT
decoder
Depth
Normals
Pose MLP
Planar pose
(θ, x, y)
Object ID → embedding
Unlabeled
real image
Initial state
q_0
$z_0 = \max(\hat{d})$
Mesh alignment
Mesh $\mathcal{M}$
Mesh depth
$d(\mathcal{M}, q_t)$
Aligner $\mathcal{A}$
ResNet-18
RGB + depth
Update state
$q \leftarrow q + \Delta q$
q_6
reproject and repeat ×6
$\mathcal{A}$ trained on perturbed states of the same K real contacts
Pseudo-labels at q_6
Depth
Normal
Planar pose (θ, x, y)
supervision on unlabeled real images
Labeled +
unlabeled real
Student f_1
same architecture as f_0
Object ID → embedding
predicted depth, normals, pose
Training loss
depth, normal, pose
Registered labels
K real contacts per object
Dashed encoder border:
frozen SITR weights.
(b) INFERENCE
Standard inference: no mesh required
Real image
Trained student f_1
SITR → DPT / Pose MLP
Known object ID
Depth
Normals
Planar pose
(θ, x, y)
Optional mesh-assisted pose refinement
Known mesh $\mathcal{M}$
Initial state
pose + $\max(\hat{d})$
Mesh projection + aligner $\mathcal{A}$
same aligner weights; six updates
Same real image
Refined planar pose
(θ, x, y)